

Source problem excerpt

arXiv:1312.3798, *The periodic decomposition problem*

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The following is transcribed from the cached original TeX, lines 491–499. The displayed source has a harmless index typo $\|x_n\|$ where the surrounding text and maximum over j require $\|x_j\|$.

Find the best constant C_n such that any $x \in \ker(T_1 - I) \cdots (T_n - I)$ has some decomposition $x = x_1 + \cdots + x_n$, with $x_j \in \ker(T_j - I)$, and

$$\max_{j=1,\dots,n} \|x_j\| \leq C_n \|x\|.$$

We saw $C_n \leq 2^{n-1}$ in general and $C_n \leq 2^{n-2}$ for translations on $BC(\mathbb{R})$. Are these estimates sharp? Is it true that $C_n = 1$ for translations on $BC(\mathbb{R})$ for every n ? Under which conditions on E and/or $T_1, \dots, T_n$ does $C_n = 1$ hold?

The source immediately before this problem states that the translation bound is sharp for $n = 2$, hence the known two-period constant is exactly one.